Noah Wilson

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Robotics • Mechatronics • Dexterous Manipulation • Robot Learning

EDUCATION

University of Florida, Gainesville, FL
B.S. Mechanical Engineering, Honors Program — Expected May 2028
GPA: 3.66/4.00 | Dean's List: Fall 2025, Spring 2026

EXPERIENCE

AeroVironment | Mechanical Engineering Intern — Sunrise, FL | Jun 2026 – Aug 2026

- Redesigned a production electronics test fixture using tolerance stack-up analysis and a custom 3D-printed base, integrating power delivery, PCB mounting, hardware, and computer interfaces
- Developed a production alignment fixture through CAD, SolidWorks FEA, prototyping, and testing; refined tolerances and engineering drawings with manufacturing technicians
- Created a BOM covering 120+ mechanical and electrical components, verified compatibility against datasheets, and managed part numbers, vendors, and documentation in PLM/PDM

UF ICIC Lab | Undergraduate Researcher - Dexterous Robotics — Gainesville, FL | Apr 2026 – Present

- Modified and integrated a 12-DOF hand exoskeleton with tactile sensors, cameras, encoders, and embedded electronics for dexterous manipulation research
- Executed a 20-participant study across four manipulation tasks, alternating between participant data collection and real-time monitoring of the multimodal acquisition system
- Synchronized and preprocessed tactile, visual, wrist-pose, and encoder data, including 1,215 demonstrations and 204,586 aligned samples across 31 recording sessions
- Recalibrated sensors, repaired electromechanical faults, and created publication figures from processed datasets
- Coauthored two IEEE ICRA 2027 submissions on adaptive exoskeleton design and cross-hand tactile supervision
- Independently created and delivered a tactile-sensing and edge-AI poster presentation to the Office of Naval Research

UF Bio-Sensing, Automation & Intelligence Lab | Undergraduate Researcher — Gainesville, FL | Aug 2026 – Present

- Developed parametric CAD for a modular blueberry-harvesting gripper with independently yielding teeth, interchangeable cartridges, and integrated collection channels; evaluated packaging and assembly clearances
- Developed a Python/MuJoCo simulator of compliant stems, spring-loaded teeth, fruit contact, and attachment release; used force and motion diagnostics to refine retaining geometry
- Restored a failed DYNAMIXEL gripper by redesigning its servo-to-gear connection; developed Python controls for motion limits and speed

PROJECTS

Wearable Robotic Third Thumb | Independent Project — Jul 2026 – Present

- Designed and built a tendon-driven wearable robotic thumb with custom 3D-printed structures, dual servos, ESP32 control, Bowden cables, and soldered circuitry; tested closure under load
- Prototyping a second-generation mechanism with independent yaw and curl, spring-loaded cam returns, bearing-supported joints, and opposing yaw tendons to address first-generation motion limitations
- Iterated joint housings, shaft supports, fastener interfaces, and cable routing in response to physical assembly failures; used parametric CAD, sampled motion-clearance checks, and print-layer inspection to refine manufacturability

Robot Learning in Simulation | Independent Project — May 2026 – Aug 2026

- Built an AWS robot-learning pipeline with MuJoCo Playground, JAX, and Brax PPO; trained a Unitree G1 locomotion policy for 202M+ simulated transitions
- Developed a Panda manipulation curriculum achieving 85% stable placement across 60 randomized simulation episodes and three evaluation seeds

TECHNICAL SKILLS

- **CAD & Manufacturing:** SolidWorks (CSWA), build123d, Fusion 360, Onshape, GD&T, FEA, Additive Manufacturing
- **Hardware:** Sensor Integration, Tactile Sensing, DYNAMIXEL, ESP32, STM32, Embedded Systems, Soldering
- **Programming & Software:** Python, C/C++, ROS 2, Linux, Git, NumPy, Pandas, Java
- **Robot Learning:** MuJoCo, Reinforcement Learning, PyTorch, JAX, Brax, PPO, AWS EC2